



5.3 Optimisation and related rates

Calculus · IB Diploma · Concept guide and original exam-style practice

Understand the concept

To find maximum/minimum values of a function, solve $f'(x)=0$ for critical points, then use the second derivative test ($f''(x)>0$ local minimum, $f''(x)<0$ local maximum) or a sign change in $f'(x)$ to classify them. Constrained optimisation problems use a given relationship to write the quantity to be optimised as a function of one variable before differentiating. Related rates problems connect several changing quantities via the chain rule, e.g. $dV/dt = dV/dr \cdot dr/dt$.

Worked example: follow the reasoning

A rectangle has perimeter 40 m. Find the dimensions that maximise its area.
Let width= x , length= $20-x$ (since $2x+2l=40$). Area $A=x(20-x)=20x-x^2$.
 $dA/dx=20-2x=0 \rightarrow x=10$.
 $d^2A/dx^2=-2<0$, confirming maximum. Dimensions: 10 m by 10 m (a square), max area=100 m².

Quick check · multiple choice

For $f(x)=x^3-3x$, the critical points occur where $f'(x)=0$. What are they?

- A $x=0$ only
- B $x=1$ and $x=-1$
- C $x=3$ and $x=-3$
- D $x=0$ and $x=3$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A cylindrical can (open top) is to have volume 500 cm³. Let r be the base radius. [9]
(a) Show that the surface area (base + curved side, no top) is given by $S = \pi r^2 + 1000/r$.
(b) Find the value of r that minimises S , and confirm it is a minimum.
(c) A spherical balloon is being inflated so that its volume increases at a constant rate of 20 cm³/s. Find the rate of change of its radius when $r=5$ cm.

Final answer: _____



Answers and explanation

Question 1 [9]

(a) $V = \pi r^2 h = 500 \rightarrow h = 500/(\pi r^2)$. $S = \pi r^2 + 2\pi r h = \pi r^2 + 2\pi r(500/(\pi r^2)) = \pi r^2 + 1000/r$.

(b) $dS/dr = 2\pi r - 1000/r^2 = 0 \rightarrow 2\pi r^3 = 1000 \rightarrow r^3 = 1000/(2\pi) = 159.15 \rightarrow r = 5.44 \text{ cm (3 s.f.)}$.

$d^2S/dr^2 = 2\pi + 2000/r^3 > 0$, confirming a minimum.

(c) $V = (4/3)\pi r^3$, $dV/dr = 4\pi r^2$. $dV/dt = dV/dr \cdot dr/dt \rightarrow 20 = 4\pi(25) \cdot dr/dt \rightarrow dr/dt = 20/(100\pi) = 1/(5\pi) \approx 0.0637 \text{ cm/s}$.

Quick check answer

B. $f'(x) = 3x^2 - 3 = 0 \rightarrow x^2 = 1 \rightarrow x = \pm 1$.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.